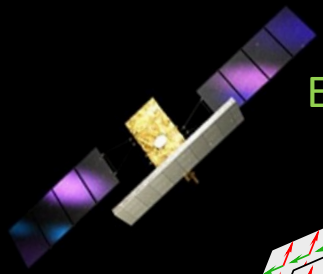
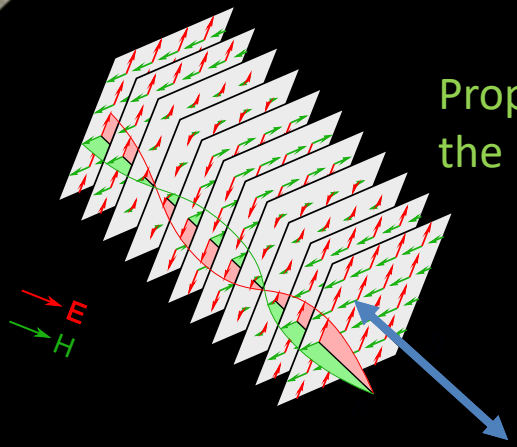
Remote sensing, plane waves, and the Fourier transform



Electromagnetic (EM) source



Propagation in the medium



Target to be imaged

Abstract

EM **Remote Sensing Systems**: layout and elementary components

EM propagation in free space: waves and the wave equation

EM propagation in the presence of sources: **impulse response and spherical waves**

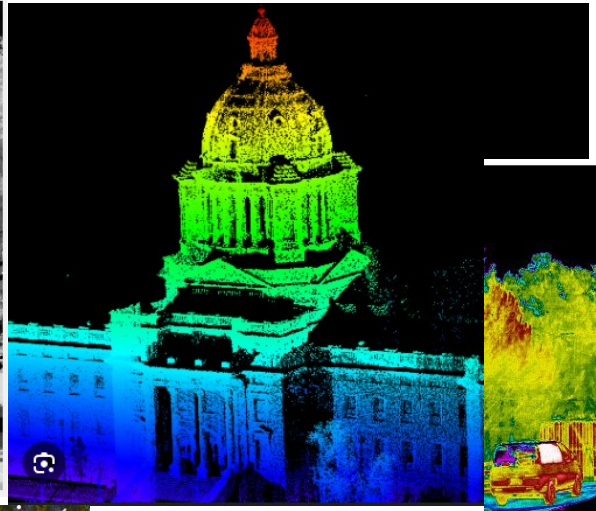
The narrowband case: **monochromatic** spherical waves

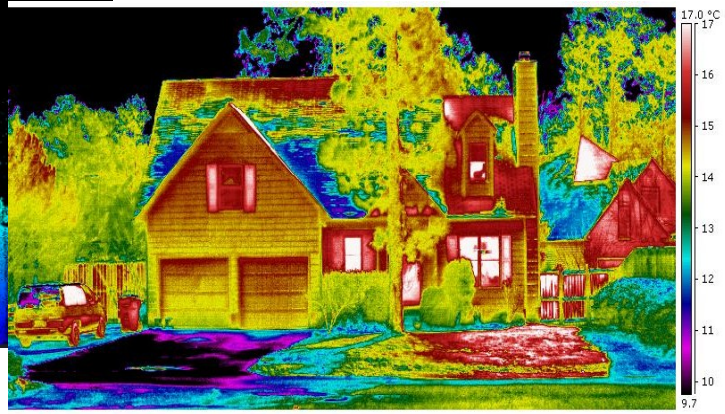
The general case: **decomposition into monochromatic components**

Why plane waves? Local approximation of spherical waves and basis for radar imaging

Classification and quality of remote-sensing images









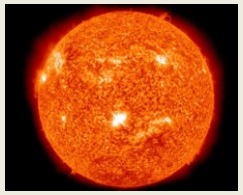
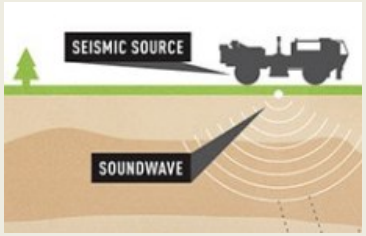
Try to classify in terms of :

active/passive,
monochromatic/panchromatic,
wavelength.

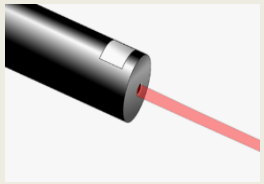
Try to qualify in terms of coverage, resolution.

Typical remote-sensing setup and waves

Source of energy




Medium
(terrain, free space, …)

Which sources are used for passive and active remote sensing?

Can you tell which are directive and which isotropic ? and which Sound waves ?

Can you tell which uses Electro Magnetic (EM) and which Sound waves ?

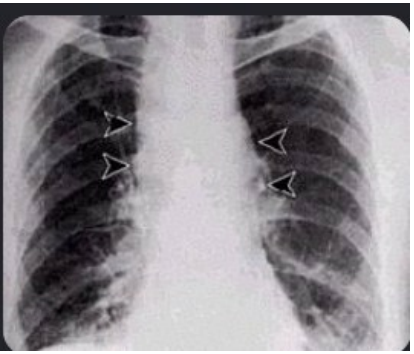
Target to be investigated

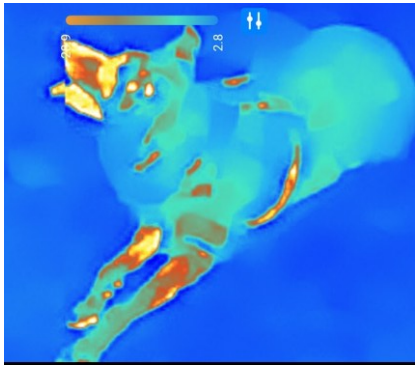


Find odd-one out

Find the odd-one out when classified as:

(1) monochromatic
(2) passive
(3) Electro-Magnetic



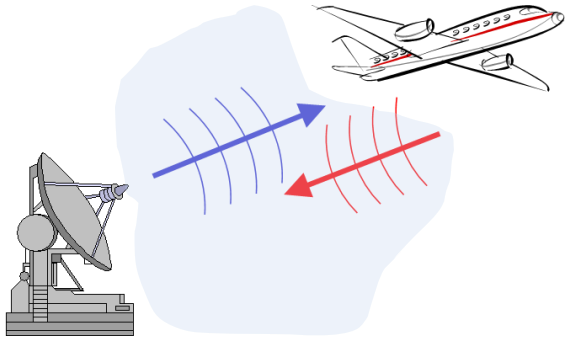






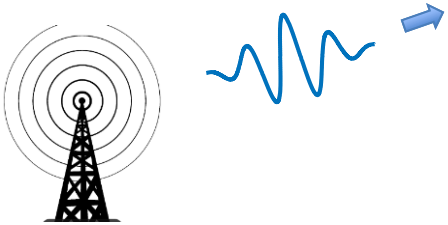
Electromagnetic remote sensing

We consider electromagnetic-field propagation in the far field.



1) The transmitter generates a time-varying EM field
E(x,y,z,t)
2) The signal travels through free space at the speed of light: v = c [m/s]
3) The signal reflects from the target
4) It propagates back and the sensor receives it

In source-free regions, propagation is described by the wave equation, a special case of Maxwell's equations, or, for harmonic fields, by the Helmholtz equation.



The source generate a signal $s(\mathbf{r}, t)$
that varies in space and time,
i.e., propagates.

Point impulse source

Questions:

1) Given the dependence on r, what can we infer about the shape?

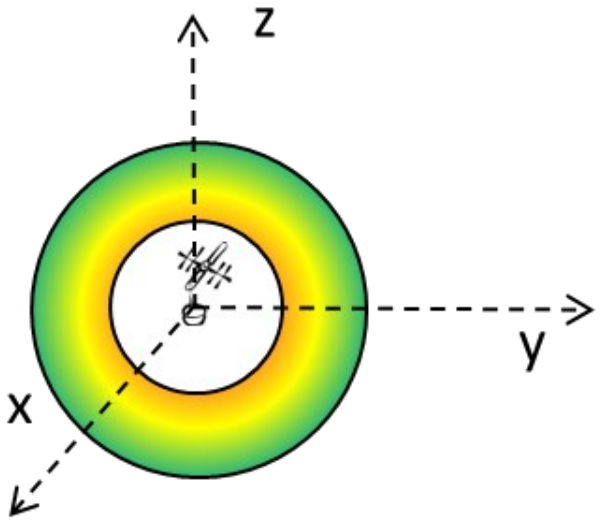
2) Does this wavefront propagate? In which direction?

3) Is it periodic?

4) Can you demonstrate that the spherical wave is the solution for impulsive input?

Spherical waves: key points

The field $E(\mathbf{r},t)=\frac{1}{r}s_p\left(t\pm\frac{r}{c}\right)$



is a spherical wave that transports the signal through 3D space. It approaches an impulse as its width tends to zero.

1) The waveform has spherical symmetry

2) For propagation, assume the '–' sign

3) The wave is not necessarily periodic

4) In the case of an exponential, introcudes a frequency and a spatial frequency and a relation between them

Electromagnetic and acoustic wave propagation in free space

$$\nabla^2 \boldsymbol{E}(\mathbf{r},t) - \frac{1}{c^2}\frac{d^2\boldsymbol{E}(\mathbf{r},t)}{dt^2} = 0$$

E is the electric field, with E = F/q [N/C].

$$\nabla^2 P(\mathbf{r},t) - \frac{1}{v^2}\frac{d^2P(\mathbf{r},t)}{dt^2} = 0$$

P is the sound-pressure field [N/m²].

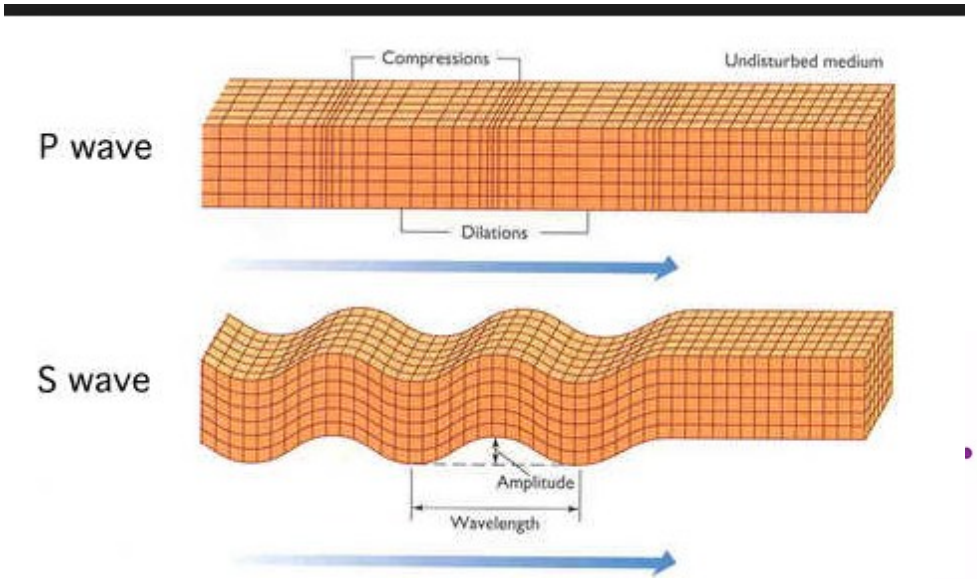
Same equation, different propagation speeds:

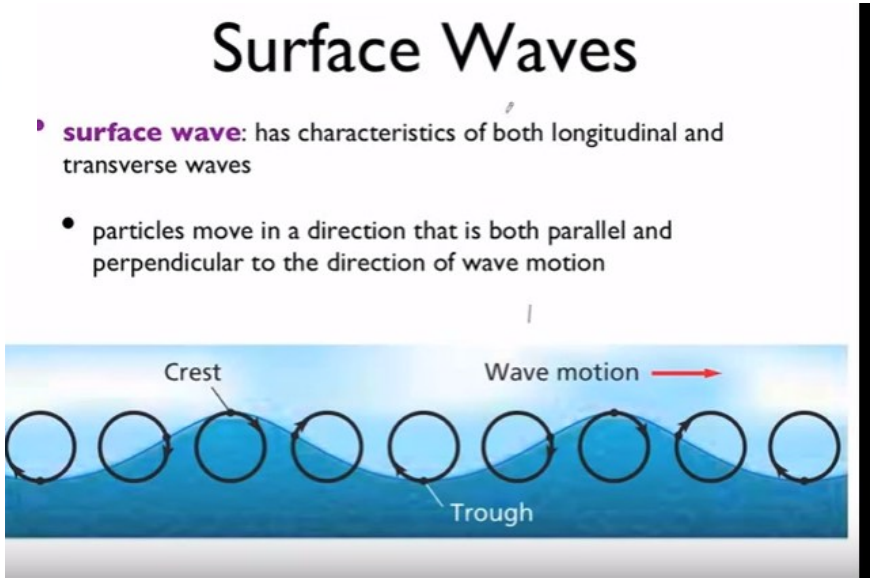
v=340 m/s in the air 20°, 1500 m/s in water, 5000 m/s in iron.
The stiffer the medium, the faster.

For EM waves in air, we commonly assume isotropic propagation.

Acoustic waves and isotropy

Acoustic propagation is often anisotropic.
It can therefore support different wave types (see Geophysical Imaging).





Spherical monochromatic wave, Fourier Transform (1D) and its interpretation

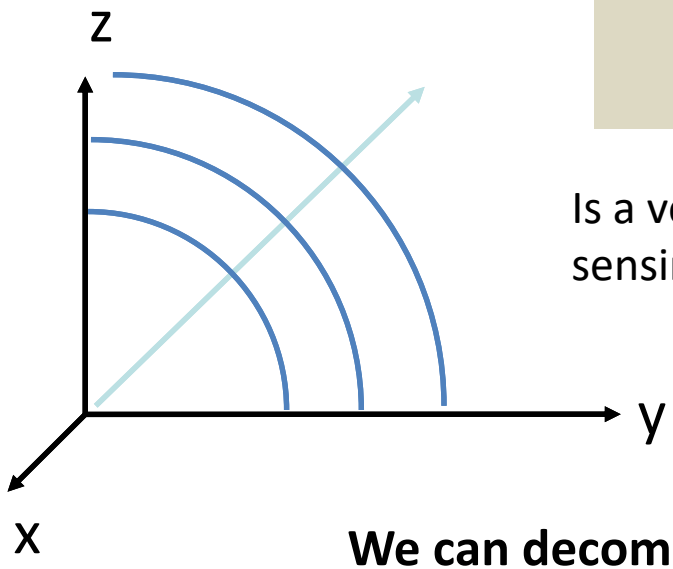
How many times we say ‘at which frequency do you transmit’ ?
This implicitly define a **monocromatic wave.**

But there is another interpretation: let us write the IFFT of a spherical wavefront

We assume the 1D case (t,r) for simplicity)

Monochromatic spherical in 3D

A monochromatic spherical wave:



$$E_{sm}(\mathbf{r},t) = \frac{1}{r} e^{j2\pi(f_0 t \pm \frac{r}{c})}$$

Is a very good approximation of many EM remote sensing (and telecom) signals.

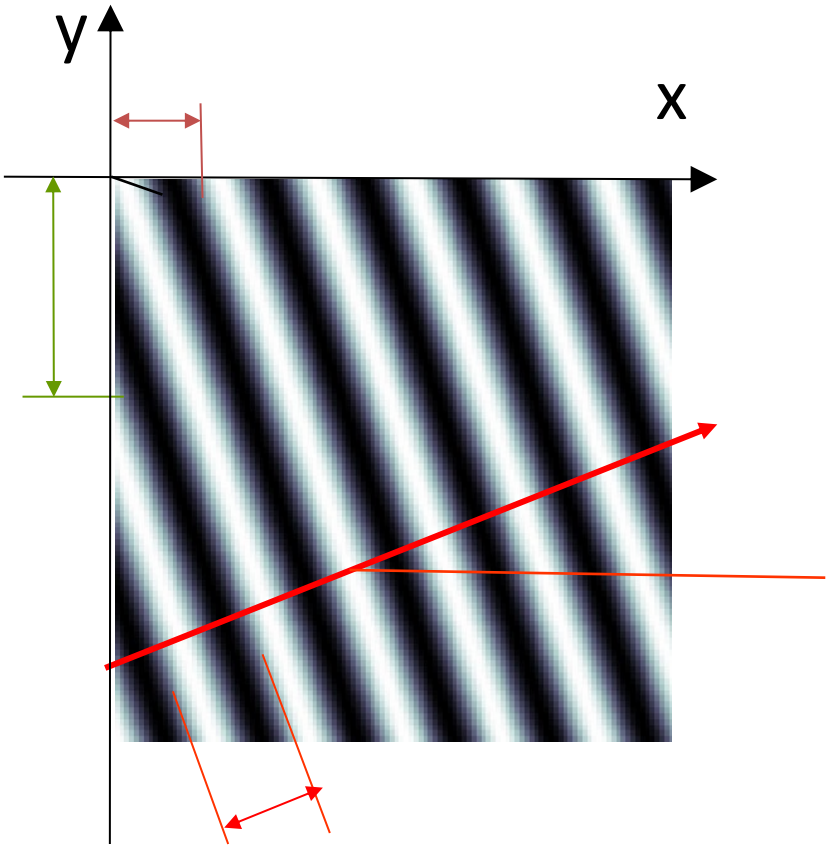
We can decompose in linear summation by the FT:

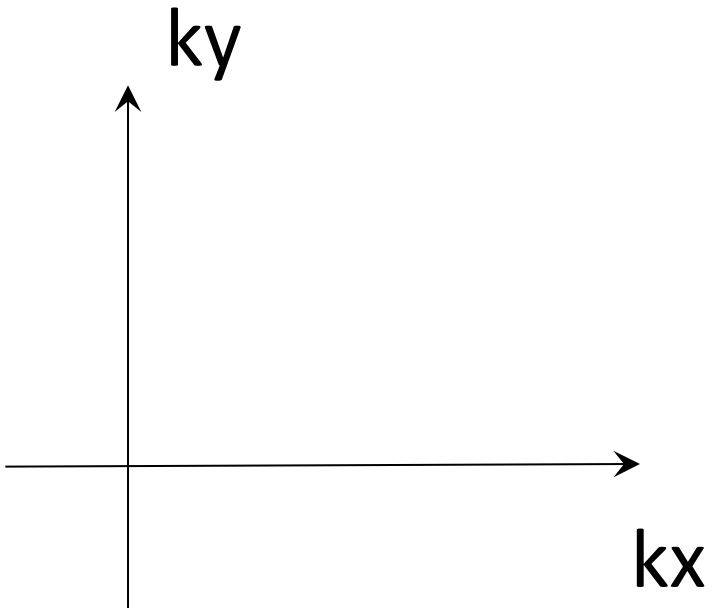
$$E_{sm}(\mathbf{r},t) = \int \iiint E(f_x, f_y, f_z, f) \cdot e^{j2\pi f t}\, e^{\pm j2\pi(f_x x + f_y y + f_z z)} df_x df_y df_z df$$

What is this signal ?

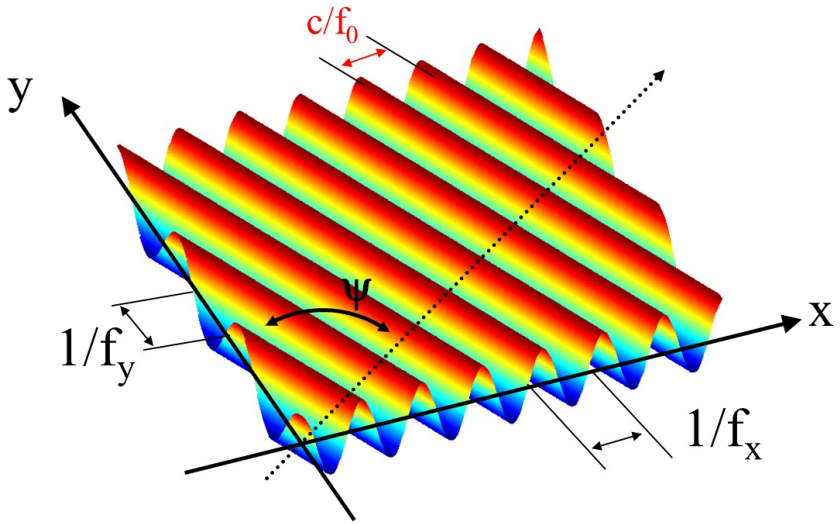
Monochromatic plane waves and their Fourier transform

$$s_{pm}(x,y,z,t) = e^{j(k_{x0}x+k_{y0}y+k_{z0}z)}e^{-j\omega_0 t}$$

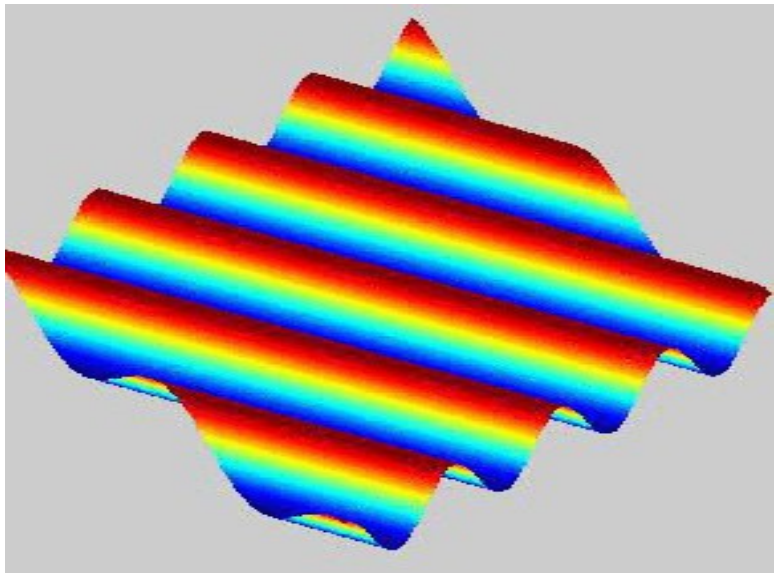




A plane wave



At a fixed time

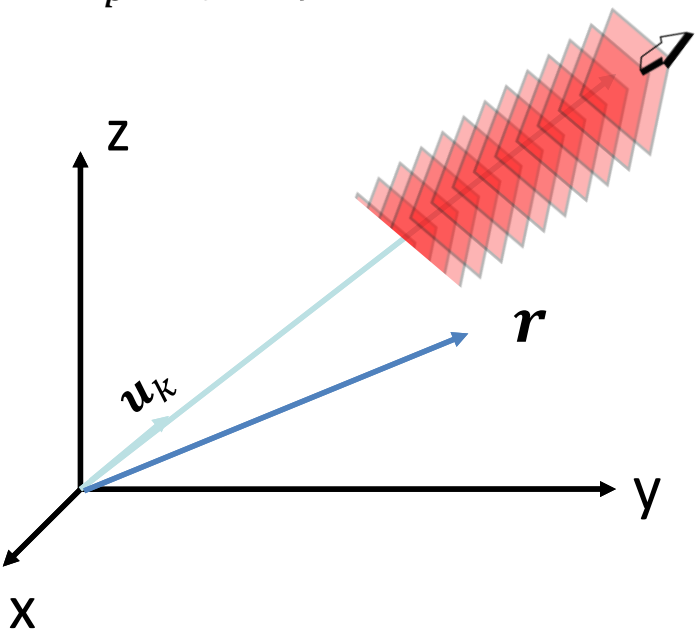


As time evolves

If this is 3D, in which direction does it propagate?

Monochromatic plane waves: the wave vector

$$s_p(x,y,z,f) = A \cdot e^{-j\frac{\omega}{c}t}\, e^{\pm j(k_x x + k_y y + k_z z)} = A \cdot e^{-j\frac{\omega}{c}t}\, e^{\pm j\boldsymbol{k}\cdot\boldsymbol{r}}$$



$$\boldsymbol{k} = \begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} \qquad \boldsymbol{r} = \begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix}$$

The plane wave is totally characterized by its wave vector: **k** that has three components.

If I evaluate sp along any line, versor **u**l, I get a sinusoid with wavenumber

$$k_l = \frac{2\pi}{\lambda_l} = \boldsymbol{k} \cdot \boldsymbol{u}_l$$

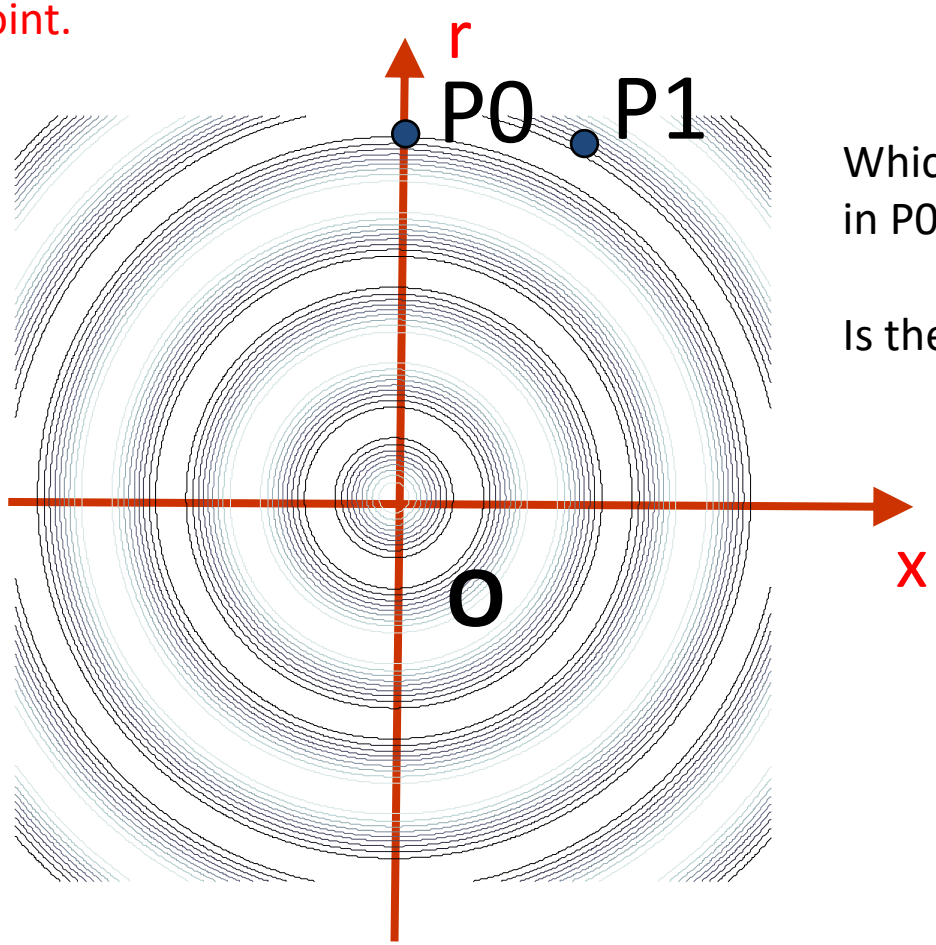
The shortest wavelength (highest spatial frequency) is measured in the propagation direction

$$\boldsymbol{k} \cdot \boldsymbol{u}_k = |\boldsymbol{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2} = \frac{2\pi}{\lambda}$$

The propagation unit vector is normal to the constant-phase plane. $\boldsymbol{u}_k = \dfrac{\boldsymbol{k}}{|\boldsymbol{k}|}$

Plane-wave approximation of a monochromatic spherical wave

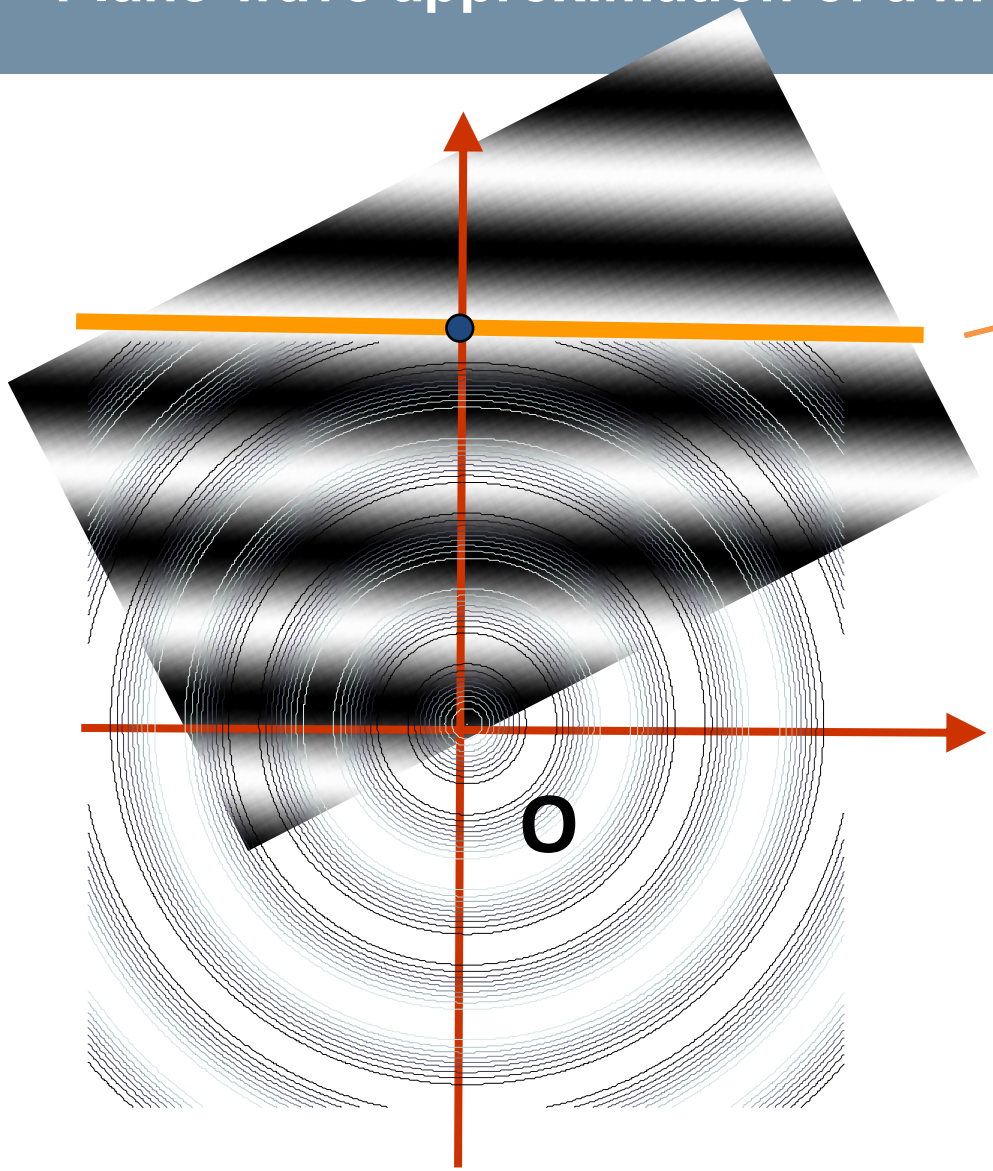
In remote sensing, a monochromatic spherical wave can be approximated by a monochromatic plane wave in the far field. The approximation depends on the observation point.

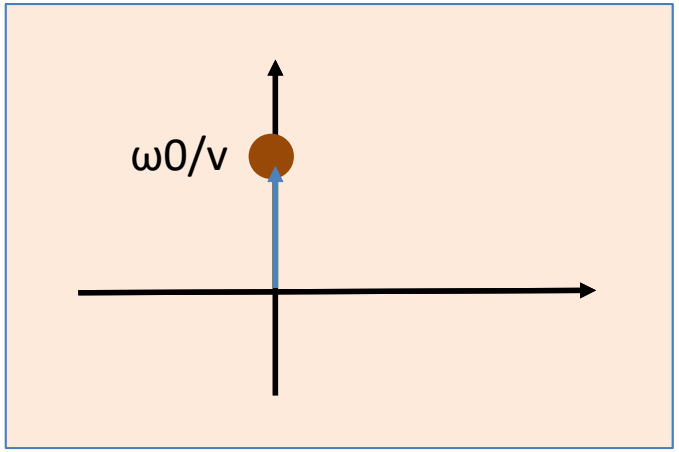


Which plane waves fit to approximate the field in P0 and P1 ?

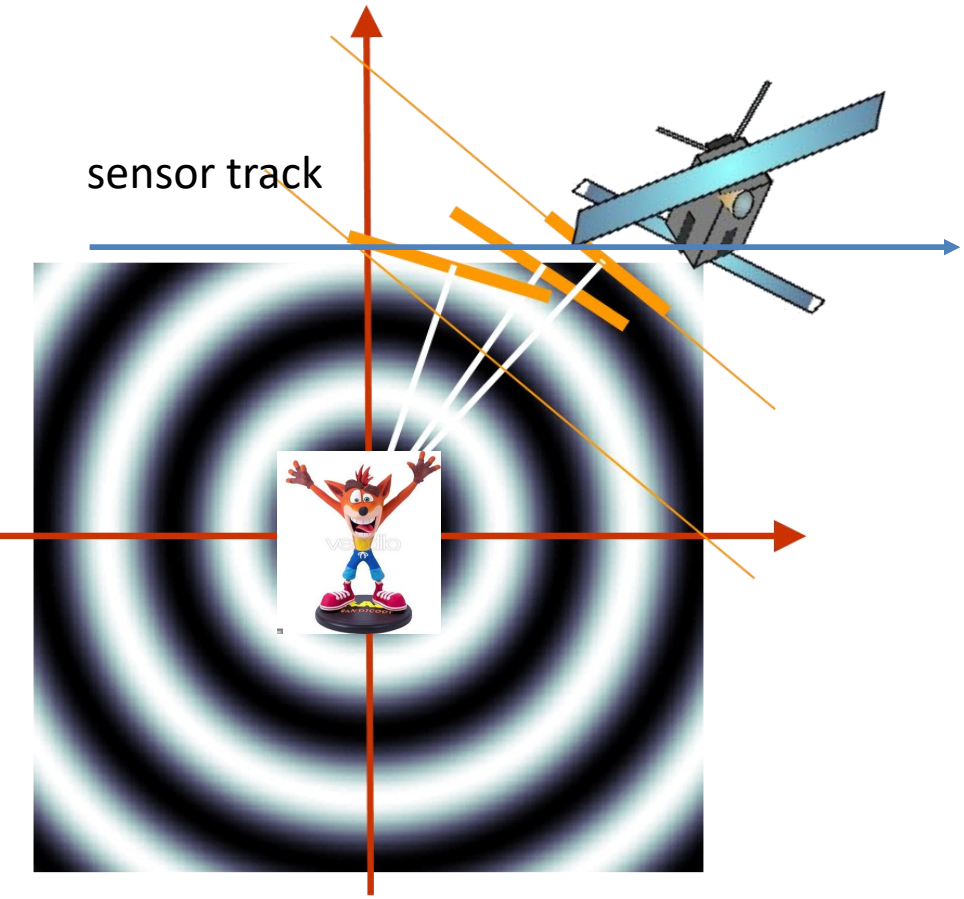
Is the plane wave a good approximation ?

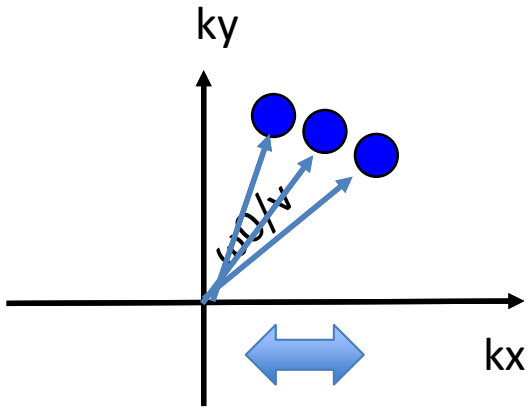
Plane-wave approximation of a monochromatic spherical wave





The synthetic-aperture concept





When the sensor observes an object from different positions along the track, it combines different k_x values. This creates a bandwidth that narrows the pulse along x.
If the entire spherical wavefront were observed, point-target resolution would be recovered.